

ELECTRONIC ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

Instruction: Using Microsoft Word/WordPad/Notepad, type Student's name, Student ID number, type Chapter title and applicable Question Number for Assignment, Mid-Term Exam or Final Exam (as listed under Brightspace, Quizzes) before typing Solution below. One separate file for each Assignment/Mid-Term/Final Exam Question.

Student's name: Amey Mahendra Thakur

Student ID number: 110107589

Assignment - RBF (3%) - Question 1:

Solution:

Input:

k	X_k	d_k
1	[0, 0]	1
2	[0, 0.9]	0
3	[1, 0]	0
4	[1, 0.9]	0

p	T_p
1	[0, 0]
2	[1, 0]

Using the Gaussian functions, $y(X_k)$ becomes

$$y(X_k) = \sum_{p=1}^2 w_p \exp(-\|X_k - T_p\|)^2 + w_0$$

To map the input to output patterns, we require,

$$y(X_k) = d_k \text{ for } k = 1, 2, 3, 4$$

Where X_k is an input vector and d_k is the desired output

The radial basis function value at the p hidden unit due to a k^{th} input pattern is equal to

$$g_{kp} = \exp(-\|X_k - T_p\|)^2 \text{ for } k = 1, 2, 3, 4; p = 1, 2$$

Repeating the above procedures, the entries of the **G** matrix can be obtained as

$$G = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \\ g_{31} & g_{32} \\ g_{41} & g_{42} \end{bmatrix} = \begin{bmatrix} 1.000000 & 0.367879 \\ 0.444585 & 0.163654 \\ 0.367879 & 1.000000 \\ 0.163654 & 0.444585 \end{bmatrix}$$

Let the combined weight vector be,

$$W_c = [w_0, w_1, w_2]^T$$

The following matrix equation can be obtained

$$W_c = H^+ D$$

where H^+ represents the pseudo-inverse of the matrix H ,

$$H = \begin{bmatrix} 1 & g_{11} & g_{12} \\ 1 & g_{21} & g_{22} \\ 1 & g_{31} & g_{32} \\ 1 & g_{41} & g_{42} \end{bmatrix} = \begin{bmatrix} 1 & 1.000000 & 0.367879 \\ 1 & 0.444585 & 0.163654 \\ 1 & 0.367879 & 1.000000 \\ 1 & 0.163654 & 0.444585 \end{bmatrix}$$

Therefore,

$$H^+ = (H^T H)^{-1} H^T = \begin{bmatrix} -0.400671 & 0.900671 & -0.400671 & 0.900671 \\ 1.318756 & -0.364698 & -0.001870 & -0.952188 \\ -0.001870 & -0.952188 & 1.318756 & -0.364697 \end{bmatrix}$$

$$W_c = H^+ D = \begin{bmatrix} -0.400671 & 0.900671 & -0.400671 & 0.900671 \\ 1.318756 & -0.364698 & -0.001870 & -0.952188 \\ -0.001870 & -0.952188 & 1.318756 & -0.364697 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$W_c = \begin{bmatrix} -0.400671 \\ 1.318756 \\ -0.001870 \end{bmatrix}$$

Therefore, the sum of the 3 weight values is **0.9162**

Matlab calculation:

The MATLAB R2023b interface displays the script `RBF_AMEY.m` in the Editor and the Command Window output.

Script Code:

```
1 format long;
2
3 % Define the matrix a
4 a = ([1 1 0.367879; 1 0.444858 0.163654; 1 0.367879 1; 1 0.163654 0.444858])
5
6
7 %Transpose the matrix
8 b = a'
9
10 c = inv(c)
11
12 Hplus = d*b
13 sum(w)
```

Command Window Output:

```
New to MATLAB? See resources for Getting Started.
c =
    4.000000000000000    1.976391000000000    1.976391000000000
    1.976391000000000    1.360016230521000    0.881363582264000
    1.976391000000000    0.881363582264000    1.360016230521000

d =
    1.943488702819009   -1.713718290377773   -1.713718290377773
   -1.713718290377773    2.778788368327468    0.689590693058206
   -1.713718290377773    0.689590693058206    2.778788368327468

Hplus =
   -0.400670558504649    0.900670558504649   -0.400670558504649    0.900670558504649
    1.318756012521255   -0.364697779138605   -0.001869711167627   -0.952188522215023
   -0.001869711167627   -0.952188522215023    1.318756012521255   -0.364697779138605

w =
   -0.400670558504649
    1.318756012521255
   -0.001869711167627

b =
    0.916215742848979
```

A red box highlights the value of `b` as `0.916215742848979`.